Then the map f will be called a **line in the horizontal foliation** of ϕ. If L is the image of a lift of f to the universal cover $\tilde{S}$ then we will refer to L as a **line** as well.

Note that lines $L \subset \tilde{S}$ can be characterized as follows. Let $\tilde{\phi}$ denote the lift of ϕ to $\tilde{S}$. Let $\tilde{S}_+, \tilde{S}_-$ be the two connected components of $\tilde{S} - L$. Then either $\tilde{S}_+$ or $\tilde{S}_-$ is disjoint from the critical trajectory of $\tilde{\phi}$ which contains L. It is clear that if $f_n : \mathbb{R} \to L_n$ is a sequence of lines in $\tilde{S}$ which converge uniformly on compacts to a map f_∞ then f_∞ is again a line.

If S is compact then the group $F = \pi_1(S)$ acts on $(\tilde{S}, \tilde{d}_\rho)$ isometrically with compact quotient. Identify $\tilde{S}$ with $\mathbb{H}^2$. Thus $\tilde{d}_\rho$ is quasi-isometric to the hyperbolic metric (see §3.9), in particular, lifts of lines of ϕ to $\mathbb{H}^2$ are quasigeodesics of the hyperbolic metric. We conclude that there is a canonical homeomorphism between the ideal boundary $\partial_\infty \mathbb{H}^2$ (with respect to the hyperbolic metric) and $\partial_\infty \tilde{S}$ (with respect to the metric $\tilde{d}_\rho$).

5.4. Teichmüller spaces of Riemann surfaces

Suppose that S, S' are connected Riemann surfaces of finite type, $\psi : \pi_1(S) \to \pi_1(S')$ is an isomorphism. This isomorphism is said to *preserve the peripheral structure of these surfaces* if for any element $\gamma \in \pi_1(S)$ corresponding to a puncture of S, its image $\psi(\gamma)$ corresponds to a puncture of S'. If S, S' are oriented then we call an isomorphism $\psi : S \to S'$ *admissible* if the induced isomorphism $H_2(S, \partial S; \mathbb{Z}) \to H_2(S', \partial S'; \mathbb{Z})$ sends the fundamental class to the fundamental class and ψ preserves the peripheral structure. Suppose that $S = \mathbb{H}^2/F, S' = \mathbb{H}^2/F'$ are hyperbolic surfaces where $F, F' \subset PSL(2, \mathbb{R})$ are Fuchsian groups. An isomorphism $\psi : F \to F'$ corresponds to an admissible isomorphism of the fundamental groups provided that it is *type-preserving*, i.e.: it sends parabolic elements to parabolic, hyperbolic elements to hyperbolic and respects the orientation [2].

Exercise 5.3. *A conformal automorphism of a hyperbolic Riemann surface of finite area is homotopic to the identity iff it equals the identity.*

Theorem 5.4. *(Nielsen Realization Theorem.) Any admissible isomorphism*

$$\psi : \pi_1(S) \to \pi_1(S')$$

is induced by an orientation-preserving quasiconformal homeomorphism $h : S \to S'$.

More generally:

Theorem 5.5. *Suppose that* $F, F' \subset Isom(\mathbb{H}^2)$ *are Fuchsian groups (they could have torsion), such that* $\mathbb{H}^2/F, \mathbb{H}^2/F'$ *have finite area and* $\psi : F \to F'$ *is a type-preserving isomorphism. Then there exists a* ψ*-equivariant quasiconformal homeomorphism* $f : \mathbb{H}^2 \to \mathbb{H}^2$.

[2] ψ is induced by a map $S \to S'$ which maps relative fundamental class to S to the relative fundamental class of S'.

Nielsen's theorem establishes existence of a quasiconformal homeomorphism in the given homotopy class; a theorem of Epstein below establishes the uniqueness of such homeomorphism up to isotopy. Epstein proved his theorem in the PL context, his result was generalized by Earle and McMullen in [EM88].

Theorem 5.6. *(D.B.A. Epstein, [Eps66]) Suppose that S_1, S_2 are hyperbolic Riemann surfaces of finite type and $f_0, f_1 : S_1 \to S_2$ are homotopic quasiconformal homeomorphisms. Then f_0 is isotopic to f_1 (through quasiconformal homeomorphisms).*

We will prove theorems 5.5 and 5.6 (and a generalization of the latter) in §8.4.

Fix once and for all a Riemann surface S of finite type, which has genus g and n punctures. Suppose that Y, Z are Riemann surfaces. Two quasiconformal homeomorphisms $f : S \to Y, g : S \to Z$ are *equivalent* if $f \circ g^{-1}$ is homotopic to a conformal map between Z and Y. The equivalence class is called a *marked Riemann surface* and denoted $[Y, f]$. The space of marked Riemann surfaces $[Y, f]$ is the *Teichmüller space* $\mathcal{T}(S)$ of the surface S. From now on we will consider Riemann surfaces of hyperbolic type (i.e. whose universal covers are conformal to the unit disc). If S has hyperbolic type, then there is the following equivalent description of the Teichmuller space $\mathcal{T}(S)$. Let $S = \mathbb{H}^2/F$ where F is a Fuchsian subgroup of $PSL(2, \mathbb{R})$. Consider the space $Hom_a(F, PSL(2, \mathbb{R}))$ of *admissible isomorphisms* between F and Fuchsian subgroups of $PSL(2, \mathbb{R})$. Take the quotient

$$\mathcal{T}_R(F) = Hom_a(F, PSL(2, \mathbb{R}))/PSL(2, \mathbb{R}) \subset \mathcal{R}_{par}(F, \{H_i\}; PSL(2, \mathbb{R}))$$

where H_i, $i = 1, ..., n$, are the cyclic subgroups of $F = \pi_1(S)$ generated by small simple loops around the punctures of S. We will refer to $\mathcal{T}_R(F)$ as the *Teichmüller space* of the Fuchsian group F.

Remark 5.7. L. Keen proved in [Kee71, Kee73] that there exists a collection of $6g - 6 + 2n$ elements $\gamma_j \in \Gamma$ (represented by simple loops in S) so that a point $[\rho] \in \mathcal{T}_R(\Gamma)$ is uniquely determined by the vector

$$(\ell(\rho(\gamma_1)), ..., \ell(\rho(\gamma_{6g-6+2n}))).$$

Recall that $\ell(h)$ is the translation length of an isometry $h : \mathbb{H}^2 \to \mathbb{H}^2$.

Nielsen's theorem provides a natural bijection between $\mathcal{T}(S)$ and $\mathcal{T}_R(F)$. In this approach $\mathcal{T}(S)$ is the collection of *marked* hyperbolic surfaces of finite area which are homeomorphic to S (the *marking* is given by a choice of an *admissible* isomorphism between the fundamental groups).

What today is known as the *Teichmüller space* was actually introduced by Fricke and Klein back in 1897–1912 [FK], although without reference to quasiconformal mappings (which were not yet invented). It also (citing Ahlfors) "requires an incredibly patient reader" to recognize $\mathcal{T}_R(F)$ in the 2-volume book [FK].

Teichmüller metric on $\mathcal{T}(S)$. If $p = [X, f_0], q = [Y, h_0] \in \mathcal{T}(S)$ then the *Teichmüller distance* between p and q is given by

$$d_{\mathcal{T}(S)}(p,q) = inf\{\frac{1}{2}\log K(f \circ h^{-1}) : (X,f) \in p, (Y,h) \in q\}$$

where $K(\tau)$ is the coefficient of quasiconformality of the quasiconformal homeomorphism τ between Riemann surfaces. The function $d_{\mathcal{T}(S)}$ is a complete metric in $\mathcal{T}(S)$. Any pair of distinct points is connected by a unique geodesic and $\mathcal{T}(S)$ is geodesically complete. However $d_{\mathcal{T}(S)}$ is not a Riemannian metric.

In what follows it will be important to distinguish surfaces with opposite orientations. If S is an oriented surface then $\bar{S}$ will denote the same surface with the opposite orientation. Each complex structure on S determines an orientation, thus if S is a Riemann surface, then $\bar{S}$ has canonical complex structure obtained by composing all the (local) coordinate maps $S \to \mathbb{C}$ with the complex conjugation in $\mathbb{C}$. Note that the canonical map $\mathcal{T}(S) \to \mathcal{T}(\overline{S})$ (which reverses orientation on each marked Riemann surface) is an isometry of Teichmüller spaces.

The infimum in the definition of Teichmüller distance $d_{\mathcal{T}(S)}(p,q)$ is always achieved. The corresponding *extremal* quasiconformal homeomorphism $\tau = f \circ h^{-1}$ is unique. Its Beltrami differential $\mu = \mu_\tau$ is described as follows (provided $p \neq q$). There exists a holomorphic quadratic differential φ on the surface Y (which is unique up to a positive scalar multiple), so that

$$\mu = t\bar{\varphi}/|\varphi| \tag{5.2}$$

where $t > 0$ is a constant. Quasiconformal homeomorphisms $\tau : Y \to X$ whose Beltrami differentials μ_τ are of the form (5.2) are called *Teichmüller maps*. The quadratic differential φ (which is defined up to a positive real multiple) is called the *initial* quadratic differential for τ. There is a unique (up to a positive real multiple) quadratic differential ψ on X so that:

- The mapping τ is affine (away from singular points) with respect to the Euclidean structures with conical singularities corresponding to φ and ψ.
- The mapping τ sends the horizontal foliation of φ to the horizontal foliation of ψ.

The quadratic differential ψ is called *the terminal quadratic differential of* τ.

The extremal quasiconformal homeomorphism τ is real-analytic on Y everywhere except at a finite collection of points which is always nonempty provided that Y is compact and τ is not conformal: these singular points are zeroes of φ. The mapping τ is not differentiable at these points.

Remark 5.8. If Y is not compact then τ can be real-analytic and non-conformal everywhere on Y. For instance, let T be a flat 2-torus, $A : T^2 \to$

T^2 an affine mapping which is not an isometry. Then for any $x \in T^2$ the restriction

$$A : T^2 - \{x\} \to T^2 - \{A(x)\}$$

is a Teichmüller mapping of $Y = T^2 - \{x\}$ which is not conformal.

Sometimes it will be important to think of $\mathcal{T}(S) = \mathcal{T}_R(F)$ as the quotient of $Belt(S) \cong Belt(\mathbb{H}^2, F)$. Namely, let $\mu \in Belt(\mathbb{H}^2, F)$ and $f_\mu = f : \mathbb{H}^2 \to \mathbb{H}^2$ be a homeomorphism which is a solution of the Beltrami equation $\overline{\partial} f = \mu \partial f$. Then f (which is unique up to postcomposition with a conformal automorphism of $\mathbb{H}^2$) is ρ-equivariant with respect to a monomomorphism $\rho : F \to PSL(2, \mathbb{R})$ whose image is a discrete subgroup $\rho(F) \subset PSL(2, \mathbb{R})$. The map $\mu \mapsto [\rho]$ determines a projection $\Phi : Belt(S) \to \mathcal{T}(S)$. Both $Belt(S)$ and $\mathcal{T}(S)$ are complex manifolds (although $Belt(S)$ is infinite-dimensional). The fact that normalized solution of the Beltrami equation depends smoothly on μ implies that the mapping Φ is smooth. It is easy to see that Φ is a submersion.

Actually, the space $\mathcal{T}(S)$ has a natural complex structure so that $\Phi : Belt(S) \to \mathcal{T}(S)$ is a holomorphic mapping; $\mathcal{T}(S)$ is biholomorphic to an open contractible domain in $\mathbb{C}^{3g-3+n}$. It is unknown if $\mathcal{T}(S)$ is *holomorphically contractible*, i.e. if there is a continuous family of holomorphic maps $f_t : \mathcal{T}(S) \to \mathcal{T}(S)$ so that $f_0 = id$, $f_1 = const$.

We now describe the tangent and cotangent spaces of $\mathcal{T}(S)$ at a point $[X, f]$. We first give a deformation-theoretic definition assuming for simplicity that X is a compact surface. Suppose that X_t is a smooth family of Riemann surfaces of finite type such that $X_0 = X$. More precisely, if $\{(U_\alpha, h_\alpha)\}$ is a finite conformal atlas on X then $\{(U_\alpha, h_{\alpha,t})\}$ is a conformal atlas on X_t, where each $h_{\alpha,t} : U_\alpha \to \mathbb{C}$ depends smoothly on the pair of variables (z, t), $z \in U_\alpha, t \in [-1, 1]$. The conformal atlas $\{(U_\alpha, h_{\alpha,t})\}$ corresponds to a collection of holomorphic transition mappings $g_{\alpha\beta,t}$ which depend smoothly on t. We thus differentiate these mappings at $t = 0$ and get a collection of holomorphic vector fields $\xi_{\alpha\beta}$ defined on $U_\alpha \cap U_\beta$. One then verifies that this collection is a cocycle $\xi \in Z^1(X, \Theta_X)$. The coboundaries $\xi \in B^1(X, \Theta_X)$ correspond to *trivial* deformations of the conformal structure on X, for such deformations there exists a smooth family of holomorphic maps $F_t : X \to X_t$, $F_0 = id$. Thus the tangent space $T_X(\mathcal{T}(S))$ is identified with the 1-st cohomology group $H^1(X, \Theta_X)$. The dual to this space is Q_X, the space of quadratic differentials, which is the cotangent bundle to $\mathcal{T}(S)$ at X.

Alternatively one identifies Q_X with the cotangent space to $\mathcal{T}(S)$ as follows. It suffices to treat the case $X = S$. Consider the projection $\Phi : Belt(S) \to \mathcal{T}(S)$. One then asks which $\nu \in L_\infty(S)$ belong to the kernel of the derivative of Φ at S:

$$\dot{f}[\nu] := \frac{d}{dt} f_{t\nu}|_{t=0} = d\Phi[\nu]$$

where $f_{t\nu} : \mathbb{H}^2 \to \mathbb{H}^2$ is the normalized solution of the Beltrami equation on $\mathbb{H}^2$. Then differentiating the Beltrami equation

$$\bar{\partial} f_{t\nu} = t\nu \partial f_{t\nu}$$

with respect to t at $t = 0$ we get:

$$\bar{\partial}\dot{f}[\nu] = \nu.$$

The condition that the deformation $f_{t\nu}$ of the conformal structure on S is infinitesimally trivial is that $\dot{f}[\nu]\partial_z$ descends to a vector field on S (for which we retain the notation $\dot{f}[\nu]\partial_z$). In particular, for each $\varphi(z)dz^2 \in Q_S$,

$$(\dot{f}[\nu]\partial_z)(\varphi(z)dz^2) = \dot{f}[\nu]\varphi(z)dz$$

is a $(1,0)$-differential on S whose exterior derivative is

$$d(\dot{f}[\nu]\varphi(z)dz) = \partial_{\bar{z}}\dot{f}[\nu]\varphi(z)\overline{dz}dz = \nu\varphi(z)\overline{dz}dz.$$

Therefore, by Stokes' theorem,

$$0 = \int_S \nu\varphi(z)dz\overline{dz} \quad \text{for each} \quad \varphi(z)dz^2 \in Q_S. \tag{5.3}$$

It turns out that the condition (5.3) is necessary and sufficient for $\nu \in Ker(d\Phi)_S$ (Teichmüller Lemma). Since the pairing (5.1) is nondegenerate, it follows that Q_S is dual to the cokernel of $(d\Phi)_S$, which is the tangent space $T_S(\mathcal{T}(S))$.

A section $Q_S \to L^1_{-1,1}(S)$ to the mapping $d\Phi$ is given by

$$\varphi dz^2 \mapsto \frac{\bar{\varphi}}{\rho}\overline{dz}/dz$$

where $\rho(z)dz^2$ is the hyperbolic metric on S. The differentials of the form $\frac{\varphi}{\rho}\overline{dz}/dz$ are called *harmonic* $(-1,1)$-*differentials* on S. They are harmonic Dolbeault representatives of the cohomology classes in $H^1(S,\Theta_S)$. See [Sch90b].

The *(Teichmüller) modular group* (the other name is the *mapping class group*) Mod_S is the quotient

$$Homeo_+(S)/Homeo_0(S)$$

where $Homeo_+(S)$ is the group of orientation preserving self-homeomorphisms of S and $Homeo_0(S)$ is the subgroup of homeomorphisms homotopic to id_S. The group Mod_S acts on $\mathcal{T}(S)$ by the precomposition $Mod_S \ni g : [X,f] \mapsto [X, f\circ g]$. This action is faithful except in the following few cases:

(a) S has genus 1 and one puncture, (b) S has genus 2 and no punctures, (c) S has genus zero and four punctures, (d) S has genus zero and three punctures.

In the first three cases the kernel of the action is the center of Mod_S (isomorphic to $\mathbb{Z}/2$), in the last case Mod_S is the permutation group on three symbols and $T(S)$ is a single point.

The group Mod_S is isomorphic to

$$Out_a(\pi_1(S)) = Aut_a(\pi_1(S))/Inn(\pi_1(S))$$

where $Aut_a(\pi_1(S))$ consists of *admissible* automorphisms. It is clear that Mod_S acts isometrically on $\mathcal{T}(S)$. Each orientation-reversing homeomorphism $f : S \to S$ induces an isometry $\mathcal{T}(S) \to \mathcal{T}(\bar{S})$ (note that $f : S \to \bar{S}$ preserves the orientation).

The group Mod_S acts on $\mathcal{T}(S)$ discretely, isometrically and biholomorphically. This action is not free, there is a subgroup of finite index $\Gamma \subset Mod_S$ which acts freely (see §5.6). The quotient $\mathcal{M}(S) = \mathcal{T}(S)/Mod_S$ is called the *moduli space* of complex structures on S. It can be viewed either as the space of *unmarked* complex structures of finite type on S or *unmarked hyperbolic structures of finite area* on S. Unless $\mathcal{T}(S)$ has complex dimension ≤ 1, the quotient of Mod_S by its center is the full group of isometries and the full group of holomorphic automorphisms of $\mathcal{T}(S)$, see [Roy71].

The moduli space $\mathcal{M}(S)$ is non-compact. Given $\epsilon > 0$, consider the subset $\mathcal{M}(S)_\epsilon$ which consists of hyperbolic surfaces where the length of each homotopically nontrivial nonperipheral loop is at least ϵ.

Theorem 5.9. *(Mumford Compactness Theorem.) The space $\mathcal{M}(S)_\epsilon$ is compact.*

See [Mum71] for algebro-geometric discussion and [Abi80] for the geometric proof.

5.5. The Weil-Petersson metric on $\mathcal{T}(S)$.

Besides the Teichmüller metric, the space $\mathcal{T}(S)$ has another useful metric d_{WP}, which is called the *Weil-Petersson metric*. Recall that the Weil-Petersson pairing $\langle \cdot, \cdot \rangle_{WP}$ on Q_S determines a metric on the cotangent space $Q_S \cong T_S^*(\mathcal{T}(S))$ and therefore a metric on $T_S(\mathcal{T}(S))$. One then verifies that this metric is infinitely differentiable and hence defines a distance function d_{WP} on the Teichmüller space. The following are the basic properties of this metric:

- Weil-Petersson metric is Kähler;
- d_{WP} is incomplete;
- the manifold $(\mathcal{T}(S), d_{WP})$ has negative sectional curvature (which is not bounded away from 0 and $-\infty$);
- $(\mathcal{T}(S), d_{WP})$ admits an exhaustion by compact convex subsets;
- the modular group Mod_S acts isometrically on $(\mathcal{T}(S), d_{WP})$.

See [Mas76], [Wol75], [Wol86], [Wol87], [Tro92], for definitions and proofs. It is unknown if $\mathcal{T}(S)$ admits a $CAT(0)$ metric which is Mod_S-invariant, one can show however that most mapping class groups Mod_S do not admit isometric properly discontinuous cocompact actions on $CAT(0)$-spaces, [KL96].

finite group $\langle \tilde{f} \rangle$ does not act freely on $\mathbb{H}^3$, which contradicts the fact that F acts freely on N.

Consider the total lifts of the groups $F, F^\#$ to $\mathbb{H}^3$: these lifts $\Gamma, \Gamma^\#$ are groups of diffeomorphisms of $\mathbb{H}^3$, and they are finite normal extensions of the group G. All elements of $F, F^\#$ induce nontrivial automorphisms of G. Extend both actions to $\widehat{\mathbb{C}}$: the restrictions of $\Gamma, \Gamma^\#$ to $\widehat{\mathbb{C}}$ are faithful representations $\Gamma, \Gamma^\# \to Homeo(\widehat{\mathbb{C}})$ and they have the same image. We conclude that the groups $\Gamma, \Gamma^\#$ are isomorphic. Hence the manifolds $M = N/F$ and $M^\# = N/F^\#$ have isomorphic fundamental groups ($\cong \Gamma$) and they are homeomorphic by Theorem 1.31. $\square$

8.6. Sullivan rigidity theorem

Suppose that $\Gamma \subset \mathrm{Isom}(\mathbb{H}^n)$, $n \geq 3$, is a discrete group whose action on its limit set is recurrent. Let $\Gamma' \subset \mathrm{Isom}(\mathbb{H}^n)$ be another discrete group and $f : \partial_\infty \mathbb{H}^n \to \partial_\infty \mathbb{H}^n$ be a quasiconformal homeomorphism which is conformal on $\Omega(\Gamma)$ and which conjugates Γ to Γ'. The following is a deep generalization of Mostow Rigidity Theorem which is due to D. Sullivan.

Theorem 8.37. *(Sullivan Rigidity Theorem, [Sul81a], [Ahl80], [Ota96].) Under the above conditions f is a Moebius transformation.*

Note that this theorem deals with the groups which are not necessarily finitely generated, we will use Theorem 8.37 in the proof of Theorem 13.11 in the case of infinitely generated groups.

According to Ahlfors' Theorem 4.123 every finitely generated discrete subgroup of $\mathrm{Isom}(\mathbb{H}^3)$ acts recurrently on its limit set. Thus, combining Theorem 4.123 with his Theorem 8.37, Sullivan proves

Theorem 8.38. *(D. Sullivan) Suppose that $G \subset PSL(2, \mathbb{C})$ is a finitely generated discrete group and μ is a G-invariant Beltrami differential such that $supp(\mu) \subset \Lambda(G)$. Then $\mu = 0$ almost everywhere.*

The following is one of the many characterizations of recurrent groups:

Theorem 8.39. *(D. Sullivan, [Sul81a].) The action of a discrete group $\Gamma \subset \mathrm{Isom}(\mathbb{H}^n)$ on its limit set is recurrent if and only if for some (any) convex fundamental polyhedron P for the action of Γ in $\mathbb{H}^n$, the intersection*

$$\partial_\infty P \cap \Lambda(\Gamma) \tag{8.6}$$

has zero Lebesgue measure.

We shall apply this theorem in the situation when the intersection (8.6) is countable.

8.7. Bers isomorphism

In this section we consider discrete, nonelementary, finitely generated groups $\Gamma \subset \mathrm{Isom}(\mathbb{H}^3)$ whose limit sets are connected. Equivalently, the

surface $\partial\dot{M}(\Gamma)$ is incompressible in $\dot{M}(\Gamma)$ (under the assumption that Γ is torsion-free). Most of the time we will deal with torsion-free subgroups $\Gamma \subset PSL(2,\mathbb{C})$, in any case will use only those quasiconformal homeomorphisms which preserve orientation. Define the *Teichmüller space* of the group Γ as

$$\mathcal{T}(\Gamma) = \{\rho \in Hom(\Gamma, \mathrm{Isom}(\mathbb{H}^3)) : \text{ there is a } \rho\text{-equivariant}$$
$$\text{quasiconformal homeomorphism } \quad f : \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}\}/PSL(2,\mathbb{C}).$$

If $f\Gamma f^{-1} \subset \mathrm{Isom}(\mathbb{H}^3)$ then the homomorphism f_* defined by

$$f_*(\gamma) = f \circ \gamma \circ f^{-1}, f_* : \Gamma \to \mathrm{Isom}(\mathbb{H}^3)$$

is called the *representation induced by* f. We let $[\rho]$ denote the $PSL(2,\mathbb{C})$-conjugacy class of $\rho \in Hom(\Gamma, \mathrm{Isom}(\mathbb{H}^3))$.

For each Γ-conjugacy class of maximal parabolic subgroups in Γ choose a representative P_j. Let $P = \{P_1, P_2, ...\}$ be the collection of representatives. I recall that the relative representation variety $\mathcal{R}(\Gamma, P; \mathrm{Isom}(\mathbb{H}^3))$ consists of conjugacy classes of representations which send elements of P_j to parabolic transformations. Thus $\mathcal{T}(\Gamma) \subset \mathcal{D}(\Gamma, \mathrm{Isom}(\mathbb{H}^3)) \cap \mathcal{R}(\Gamma, P; \mathrm{Isom}(\mathbb{H}^3))$. Let $L_\infty(\Omega(\Gamma))$ denote the space of complex-valued L_∞-functions on the domain $\Omega(\Gamma)$. The condition that $f_*(\Gamma) = f \circ \Gamma \circ f^{-1}$ consists of Moebius transformations can be expressed in terms of the Beltrami differential $\mu(z)$ of the quasiconformal homeomorphism f:

$$\gamma^*\mu = \mu, \quad \text{i.e. } \mu \text{ is } \Gamma - \text{invariant} \tag{8.7}$$

(see §5.2). Define

$$B_\infty(\Gamma) := \{\mu \in L_\infty(\Omega(\Gamma)) : ||\mu||_\infty < 1, \mu \text{ is } \Gamma\text{-invariant}\}.$$

This space is the open unit ball in the Banach space

$$L_\infty(\Gamma) := \{\mu \in L_\infty(\Omega(\Gamma)) : \mu \text{ is } \Gamma\text{-invariant}\}.$$

Thus we can think about $\mathcal{T}(\Gamma)$ as the quotient of the space $B_\infty(\Gamma)$ by the following equivalence relation:

$$\mu \sim \nu \text{ iff the quasiconformal homeomorphisms } f_\mu, f_\nu \text{ induce}$$
$$\text{the same element of } \mathcal{R}(\Gamma, \mathrm{Isom}(\mathbb{H}^3)).$$

We assume that $\mathcal{T}(\Gamma)$ has the topology of algebraic convergence induced from the usual topology of the algebraic variety $\mathcal{R}(\Gamma, \mathrm{Isom}(\mathbb{H}^3))$. Define the *Teichmüller distance* between the points $[\rho], [\phi] \in \mathcal{T}(\Gamma)$ as

$$d_{\mathcal{T}}(\rho, \phi) = \inf\{\frac{1}{2}\log K(f) | f : \widehat{\mathbb{C}} \to \widehat{\mathbb{C}} \text{ is a quasiconformal homeomorphism}$$
$$\text{such that } f_*(\gamma) = \phi \circ \rho^{-1}(\gamma) \text{ for each } \gamma \in \rho(\Gamma)\}.$$

From now on let we will assume that the group Γ is torsion-free: it will simplify our arguments and terminology. Let $\Sigma = \Omega(\Gamma)/\Gamma = \Sigma_1 \cup ... \cup \Sigma_n$ be

the disjoint union of connected components. Define the Teichmüller space $\mathcal{T}(\Sigma)$ as the direct product of Teichmüller spaces $\mathcal{T}(\Sigma_1) \times ... \times \mathcal{T}(\Sigma_n)$. We define the Teichmüller metric on the product is the maximum of Teichmüller metrics on components. We will use the notation $\mathcal{T}_t(\Gamma)$ for the set $\mathcal{T}(\Gamma)$ given the topology induced by the Teichmüller metric.

Theorem 8.40. *(L. Bers [Ber72].) There exists a natural homeomorphism $\beta : \mathcal{T}(\Sigma) \to \mathcal{T}_t(\Gamma)$ which is called the* **Bers identification**. *This homeomorphism is an isometry of Teichmüller metrics. The identity map $\mathcal{T}_t(\Gamma) \to \mathcal{T}(\Gamma)$ is always continuous. If Γ is geometrically finite then the identity map $\mathcal{T}_t(\Gamma) \to \mathcal{T}(\Gamma)$ is a homeomorphism.*

Remark 8.41. If we drop the assumption that $\partial \dot{M}(\Gamma)$ is incompressible then the map β is a covering map.

Proof: The proof which we present here depends upon the Ahlfors' Finiteness Theorem. However in §8.14 we will use arguments from this proof to prove the Ahlfors' Theorem.

We can assume that the group Γ is Kleinian (otherwise the assertion follows from the Sullivan Rigidity Theorem). Let X denote the surface Σ regarded as the origin of the Teichmüller space. Let $[f : X \to Y] \in \mathcal{T}(\Sigma)$, where f is a quasiconformal map. Then $\bar{\partial} f / \partial f$ is a Beltrami differential μ on X. Lift μ to a differential $\tilde{\mu}$ on $\Omega(\Gamma)$ using the covering $\Omega(\Gamma) \to X$. Then $\tilde{\mu}$ is invariant under Γ and we extend $\tilde{\mu}$ by zero to the limit set of Γ. The result is a Beltrami differential on $\widehat{\mathbb{C}}$ which is Γ-invariant (we retain the name $\tilde{\mu}$ for this differential) and we solve the Beltrami equation

$$\bar{\partial} h / \partial h = \tilde{\mu}. \tag{8.8}$$

The quasiconformal homeomorphism h conjugates Γ to a Moebius group Γ'. Therefore, the induced representation $\rho = h_*$ of the group Γ determines an element of $\mathcal{T}(\Gamma)$. We define the map β by the formula:

$$\beta[f : X \to Y] := [\rho].$$

The right inverse to the map β is defined as follows. Suppose that $[\rho] \in \mathcal{T}(\Gamma)$, then we let $\Gamma' = \rho(\Gamma)$, $Y = \Omega(\Gamma')/\Gamma'$ and the *marking* $X \to Y$ is obtained by descending a ρ-equivariant mapping f from $\Omega(\Gamma) \to \Omega(\Gamma')$ to $X \to Y$.

Let us check that the correspondence β is well-defined. Suppose that

$$[f : X \to Y] = [g : X \to Z]$$

is a point in the Teichmüller space of X. Then there exists a biholomorphic map $\alpha : Z \to Y$ which is homotopic to $f \circ g^{-1}$. Define the representations ρ' and ρ'' using $\beta(f)$ and $\beta(g)$. The corresponding groups Γ' and Γ'' are finitely generated and have the property that the representation $\rho' \circ (\rho'')^{-1}$ is induced by a quasiconformal map $\tilde{\alpha}$ which is conformal on $\Omega(\Gamma'')$. By Sullivan's rigidity theorem, the map $\tilde{\alpha}$ is Moebius. Thus $[\rho'] = [\rho'']$. The fact that β is onto follows from existence of the right inverse to $[\beta]$. Then

we need to check that β is injective. Suppose that $\beta[X, id] = \beta[f : X \to Y]$. Let $\tilde{\mu}$ be the Γ-invariant Beltrami differential in $\widehat{\mathbb{C}}$ obtained by lifting the Beltrami differential of f. As above, we let h denote the normalized solution of the Beltrami equation (8.8). The assumption $\beta[X, id] = \beta[f : X \to Y]$ is equivalent to saying that the representation $\rho : \Gamma \to PSL(2, \mathbb{C})$ induced by h is the identity. In particular, $\rho(\Gamma) = \Gamma$ and the Riemann surface Y is the quotient $\Omega(\Gamma)/\Gamma$, which equals X. The marking $f : X \to Y$ has to be homotopic to the identity since $\rho : \Gamma \to \Gamma$ is the identity automorphism and each component of $\Omega(\Gamma)$ is simply-connected. We conclude that $[X, id] = [f : X \to Y]$, which implies that β is injective.

Therefore β is a bijection.

Remark 8.42. We note that if X does not have finite conformal type, one has to modify the definition of the Teichmüller space of X to get a meaningful version of Bers' theorem. With the modified definition the Teichmüller space of the open unit disk is no longer a single point, it is the quotient of the quasi-Moebius automorphisms of the unit circle by the action of $PSL(2, \mathbb{R})$. Then the proof of injectivity of β becomes much more complicated.

Sullivan rigidity theorem implies now that β is an isometry of the Teichmüller metrics on $\mathcal{T}(\Gamma)$ and $\mathcal{T}(X)$. This proves the first part of the theorem.

The fact that $id : \mathcal{T}_t(\Gamma) \to \mathcal{T}(\Gamma)$ is continuous follows from continuity of solutions of the Beltrami equation with respect to the Beltrami differentials.

The bicontinuity of this map in the case of geometrically finite groups is less obvious. One way to prove it is to establish that the relative representation variety $\mathcal{R}(\Gamma, P; \mathrm{Isom}(\mathbb{H}^3))$ is smooth near any point $[\rho] \in \mathcal{T}(\Gamma)$ and has the same dimension as $\mathcal{T}(X)$. (We shall prove it in the Sections 8.8, 8.9.) Then continuity of β^{-1} follows from the invariance of domain theorem. The other way is to establish so called *quasiconformal stability* of geometrically finite groups using direct geometric arguments as it is done in [Mar74], [Sul85b]. □

Remark 8.43. If $\Gamma \subset PSL(2, \mathbb{C})$, then Σ is orientable; in this case the Teichmüller spaces $\mathcal{T}(\Gamma) \subset \mathcal{R}(\Gamma, P; PSL(2, \mathbb{C}))$ and $\mathcal{T}(\Sigma)$ have natural complex structures. One can prove that β is biholomorphic.

8.8. Smoothness of representation varieties

Let $\Gamma \subset PSL(2, \mathbb{C})$ be a finitely generated torsion-free discrete nonelementary subgroup, $M = M(\Gamma) = \mathbb{H}^3/\Gamma$. Choose $\epsilon > 0$ smaller than the Margulis constant, then M^0_ϵ is the union of the ϵ-thick part $M_{[\epsilon,\infty)}$ and of the Margulis tubes in M. Let C denote $cl(M - M^0_\epsilon)$, the union of Margulis cusps in M. Let α denote the total number of cusps and τ denote the number of rank 2 cusps in M. Choose a compact core $N := M_c \subset M^0_\epsilon$ as in Section 4.23, $P := M_c \cap C$ is the disjoint union of annuli and tori so that for each component $C_i \subset C$ the inclusion $P \cap C_i \hookrightarrow C_i$ is a homotopy-equivalence, (M_c, P) is a *pared manifold*. For each toral component $T_i \subset P$ choose an